Ship Design and Construction

Starships come in many shapes, configurations, and sizes. Players will typically operate a small vessel, though larger-scale naval, exploration, or merchant-marine campaigns are also possible. This chapter presents a system for designing ships from scratch and for upgrading existing ships.

These rules classify ships by their volume rather than mass, for reasons of tradition and since the use of the magneto-gravitic maneuver drive makes volume a more important design consideration than mass. Ship designs note vessel sizes in displacement tons: this is how many tons of liquid hydrogen the ship displaces. 1 displacement ton (d-ton or ton, for short) equals approximately 14 cubic meters: a cube roughly 2.4m per side. On a square grid map where each square is 1.5m per side and 3.0m high, each ton is represented by approximately two squares on the deck plan.

A note on tech levels: the default interstellar space-opera tech level in Cepheus Deluxe is TL12. Ships designed at lower or higher Tech Levels are possible.

Also note that these rules present interstellar-level spacecraft designs at TLs 9-16, which are ideal for space-opera campaigns. Societies at TLs 6-8 can also manufacture spacecraft for intra-system travel, but these are considerably more primitive than the interstellar-era craft one can create with these ship construction rules. For the sake of brevity, we will not describe primitive craft here.

Ship Design Checklist

1. Choose hull size.
    1. Choose hull configuration.
    2. Add armor, if desired.
2. Choose and calculate maneuver drive, jump drive, and power plant.
3. For jump-capable ships, determine fuel requirements.
4. Add bridge.
5. Choose electronics.
6. Optionally, add weaponry and shields.
7. Add additional components and carried craft.
8. Determine crew and passengers.
    1. Add staterooms and cryotubes accordingly.
9. Allocate remaining space to cargo.
10. Calculate final cost and construction time.
11. Mass-produced ships enjoy a 10% discount.



SHIP HULL

Hulls cost MCr0.1 per ton. This is modified by the ship's configuration, as given below. The largest available starship hull in Cepheus Deluxe is 10000 tons.

Starships can be constructed at Class-A starports, system ships (which lack jump drives) and small craft (vessels under 100 tons) can be constructed at Class-A or Class-B starports. Use the following table to determine the construction time of a spacecraft.

Spacecraft Construction Times

Ship Tonnage	Construction Time (weeks)
<100 tons	26
100 – 500 tons	52
501 – 1000 tons	78
1001 – 2000 tons	120
2001 – 3000 tons	150
3001 – 4000 tons	180
4001 – 5000 tons	210
5001 – 10000 tons	300

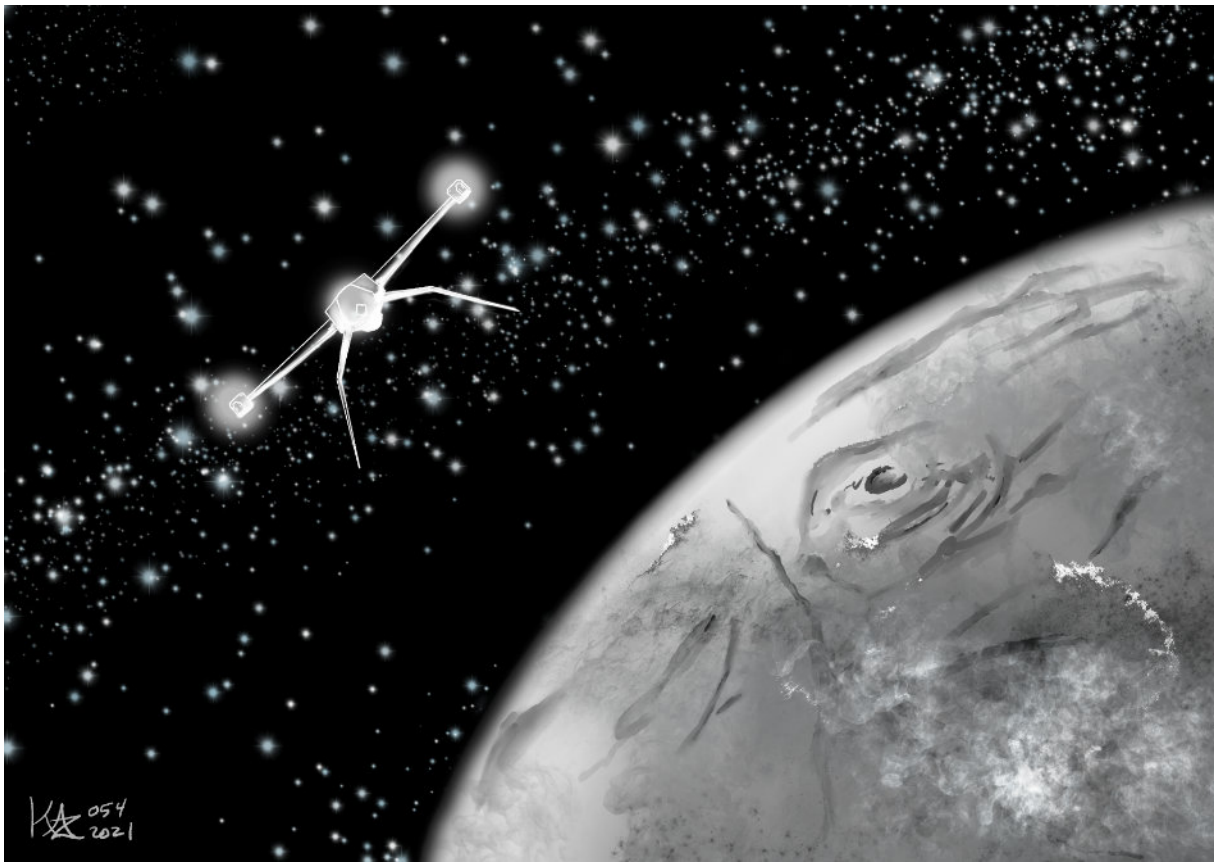
HULL CONFIGURATION

There are three general starship hull configurations:

Standard: The ship is robust enough to enter an atmosphere but will maneuver very poorly. It cannot fly like an aircraft but only land and take off into space again. The ship may skim gas giants for fuel. No additional cost is required. All Pilot skill checks when flying in an atmosphere suffer a DM-2.

Streamlined: The ship will operate and maneuver like an aircraft in an atmosphere. The ship may skim gas giants for fuel and comes equipped with Fuel Scoops at no extra cost. Streamlined hulls cost 10% more than the standard price (MCr0.11 per ton).

Distributed: The hull is designed with only space operations in mind. The ship is non-aerodynamic and will break up if it enters an atmosphere denser than Trace. It may not skim gas giants for fuel; ships with distributed hulls will often carry small craft to interface with worlds. The hull costs 10% less than the standard price (MCr0.09 per ton).



SHIP ARMOR

Ships armor is available as Light Armor and Heavy Armor. Armor affects the performance of anti-ship weapons and may even prevent damage completely from lighter weapons.

Ships of 1000 tons or more automatically increase armor by one increment: treat an unarmored 1000+ ton ship as having Light Armor, a lightly armored 1000+ ton ship as having Heavy Armor, and a heavily armored 1000+ ton ship as having Massive Armor (a category only available for such large ships).

LIGHT ARMOR

The upper table lists how many tons of space Light Armor occupies and how much it costs by Tech Level.

HEAVY ARMOR

The lower table lists how many tons of space Heavy Armor occupies and how much it costs by Tech Level.

Light Armor Technology

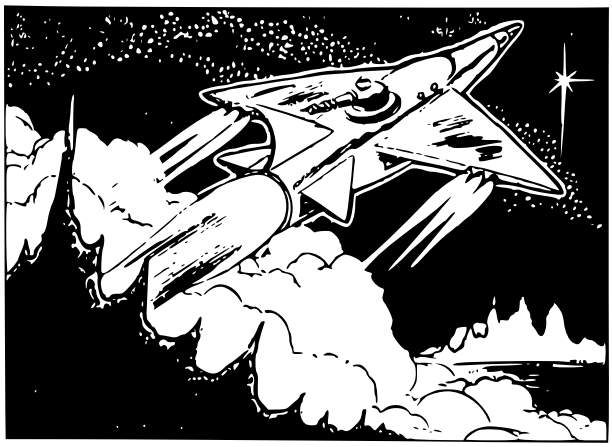
Armor Type	Tech Level	Tonnage	Cost
Titanium Steel	7	12.5% of ship tonnage	12.5% of base hull cost
Composite Alloy	8	10% of ship tonnage	20% of base hull cost
Crystaliron	10	6.5% of ship tonnage	25% of base hull cost
Superdense	12	5% of ship tonnage	30% of base hull cost
Bonded Superdense	14	4% of ship tonnage	40% of base hull cost
Collapsed Superdense	16	3.5% of ship tonnage	50% of base hull cost

Heavy Armor Technology

Armor Type	Tech Level	Tonnage	Cost
Titanium Steel	7	25% of ship tonnage	25% of base hull cost
Composite Alloy	8	20% of ship tonnage	40% of base hull cost
Crystaliron	10	12.5% of ship tonnage	50% of base hull cost
Superdense	12	10% of ship tonnage	60% of base hull cost
Bonded Superdense	14	8.5% of ship tonnage	60% of base hull cost
Collapsed Superdense	16	7% of ship tonnage	100% of base hull cost

STEALTH

Specialized emissions-masking systems and stealth structures can render a ship difficult to detect by sensors at any range. Stealth costs MCr0.1 per ton of the ship's hull. Stealth modifications provide DM-4 to any Computer throws to detect or lock onto the ship in and out of combat.



DRIVES

A system ship must have a maneuver drive (M-Drive) and a power plant (P-Plant). A starship must have a jump drive (J-Drive) and a power planet (P-Plant); a maneuver drive may also be installed but is not required. Small craft are too small to mount jump drives.

On the Drive Performance tables, the maneuver drive potential is the Thrust number (Tn), measured in the number of G's of acceleration available. For Jump Drives, the potential is the Jump number (Jn), or Jump range in parsecs. The power plant rating code must be at least equal to either the maneuver drive or Jump drive rating, whichever is higher.

Drives use a percentage of ship's tonnage. Note that higher tech levels can produce more efficient drives.

Drive and power plant performance is not linear. Small and exceptionally large components tend to be less efficient than moderately sized units. This is especially true with small craft drives, which are significantly less efficient than those of starships (100+ tons).

Note that Performance of "0" represent station-keeping ability only, no tactical maneuvering. Often used by space stations and orbital facilities. Also note that while a space station might have Maneuver Drive 0, it must have at least Power Plant 1 to operate.

Drive Cost: Jump drive, Maneuver drive, and Power Plant components cost MCr2.5 per ton.

Small Craft Drive Performance (10–99-ton hull; Percentage of hull tonnage)

Type	0	1	2	3	4	5	6
Power Plant	-	4.5	7.8	11.4	15.0	18.9	23.1
Maneuver Drive	1.5	2.7	4.5	6.6	9.0	11.7	14.7
Jump Drive	-	-	-	-	-	-	-

Ship Drive Performance (100+ ton hull; Percentage of hull tonnage)

Type	0	1	2	3	4	5	6
Power Plant	-	1.5	2.6	3.8	5.0	6.3	7.7
Maneuver Drive	0.5	0.9	1.5	2.2	3.0	3.9	4.9
Jump Drive	-	3.0	4.5	6.5	9.0	12.0	15.5

Tech Level Adjustment for Drive Percentage

Type	TL8	TL9	TL10	TL11	TL12	TL13	TL14	TL15	TL16
Power Plant	1.9	1.7	1.6	1.5	1.0	0.9	0.8	0.7	0.6
Maneuver Drive	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.6
Jump Drive	-	2.0	1.4	1.2	1.0	0.9	0.8	0.7	0.6

JUMP TECHNOLOGY

While all magneto-gravitic maneuver drives and fusion power plants are available from TL9 on, jump technology develops more gradually, as follows. Remember that the ship's computer also determines the ship's maximum jump rating.

Jump Technology

Jump Rating	Minimum Tech Level
1	9
2	11
3	12
4	13
5	14
6	15

FUEL

Fusion power plants include their own fuel, typically Helium-3. Power plant fuel suffices for over a year; annual maintenance costs include the expense of refueling the power plant. However, jump drives require liquid hydrogen fuel for each jump. When performing a jump, a starship uses 10% of its tonnage in fuel per parsec jumped. To determine how much fuel a given jump consumes, multiply the starship's tonnage by 0.1, and then multiply that number by the number of parsecs being travelled in the jump. For example, a 100-ton starship performing Jump 2 must expend 100 x 0.1 x 2 = 20 tons of fuel to operate its jump drive.

STARSHIP BRIDGE

Any ship of 100 tons or more has a bridge based on its total tonnage. The size of the bridge is 2% of the hull tonnage and costs MCr0.5 per 100 tons of ship. This tonnage also includes avionics and communication equipment, especially in larger ships. Minimum bridge tonnage is 10 tons for a starship and 5 tons on a system ship.

SMALL CRAFT COCKPIT AND ACCOMMODATIONS

A small craft control cabin or cockpit costs MCr0.1 per 20 tons of craft. A cockpit is cramped and will be uncomfortable for more than 24 hours of continuous use. Apply a DM-1 to skill checks in those cases. A control cabin is far more comfortable for longer journeys.

In addition, small craft may also mount acceleration couches, allowing them to carry passengers for short flights of 12 hours or less. These couches accommodate a single person each, displace 0.5 tons and cost Cr25000 (MCr0.025) per person. It is possible to install staterooms in small craft as well.

Note that small craft do not require additional electronics beyond the baseline model, though military and exploration craft greatly benefit from them.

Small Craft Controls

Controls	Tonnage
1-Person Cockpit	1.5 tons
2-Person Cockpit	3 tons
1-Person Control Cabin	3 tons
2-Person Control Cabin	6 tons

ELECTRONICS

Each ship has a central computer and various electronic systems. The computer core itself is exceedingly small; even with hardening and redundancy, this mainframe is rarely larger than a small 21st century beer refrigerator. The actual tonnage includes various electronic systems: wiring, terminals, sensors, and communication systems.

Electronics are required for calculating interstellar jumps. The computer included in an electronic system grant a DM to sensor throws in and out of combat. The electronics may also assist in targeting, as noted in the Space Combat chapter.

All electronic systems are hardened against radiation and electromagnetic pulses, though this hardening is neither perfect nor complete; see the radiation attack rules in the Space Combat section.

Starships need at least Model/1 electronics for jump calculations. Small craft may have baseline computers with minimal electronics: this is included in their cockpit costs and tonnage. However, baseline electronics are recommended only for civilian craft.

You may add an “Advanced” (adv) option to any electronics system. It increases the electronics cost by 50% but also permits the ship to have a maximum Jump rating one higher than what it would attain with standard electronics. For example, Model/1adv electronics would have a Sensor DM-2, cost MCr1.5, and take up 1 ton of space on board, but would allow a maximum of Jump-2.

Spacecraft Electronics

Electronics	TL	Cost	Tonnage	Max Jump Rating	Sensor DM
Baseline	-	-	-	-	DM-4
Model/1	7	MCr1	1	Jump 1	DM-2
Model/2	9	MCr3	2	Jump 2	DM+0
Model/3	11	MCr5	3	Jump 3	DM+1
Model/4	12	MCr10	5	Jump 4	DM+2
Model/5	13	MCr20	6	Jump 5	DM+2
Model/6	14	MCr40	8	Jump 6	DM+3
Model/7	15	MCr50	10	Jump 6	DM+2
Model/8	15	MCr100	12	Jump 6	DM+5

Armament

Military craft, as well as civilian craft operating on the frontier, often carry weaponry. Each weapon requires one or more hardpoints to mount on a ship. Ships have one hardpoint available per 100 tons of hull size. Fire control is included in the turret or bay tonnage amounts listed below.

Ship weapons have the following stats:

Weapon: weapon's name.

TL: minimum tech level to construct the weapon.

Type: the weapon's type, whether it is Light, Intermediate, Heavy, or a Main Gun.

Hits: the number of internal damage rolls the weapon inflicts on the appropriate damage table. Radiation damage rolls apply regardless of armor rating unless the ship is completely invulnerable to that weapon's class.

Turrets

One turret may be attached to each hardpoint on the ship.

Turret Displacement and Cost

Weapon	Tech Level	Tons	Cost (Mcr)
Single Turret	7	2	2
Double Turret	8	3	3
Triple Turret	9	4	4
Pop-Up Turret	10	x2	x2
Fixed Mounting	-	-	X0.5

Single, Double and Triple turrets can hold one, two, or three weapons, respectively.

Pop-Up is a quality that can be applied to any type of turret. The turret is concealed in a pod or recessed on the hull and is detectable only when deployed. A ship with all its weapons in pop-up turrets looks unarmed to a casual sensor scan. Pop-up turrets take up twice as much tonnage as their normal counterparts and double the turret's cost.

Fixed Mounting weapons cannot move and are limited to firing in one direction: usually fixed forward, or fixed rear. Fixed mountings are found mainly on small craft. A fixed mounting costs half as much as a turret with the same number of weapons. See the Space Combat chapter for rules on how to fire Fixed weapons.

Turrets and fixed mounts can accommodate a variety of weapon systems. The legality and availability of a given weapon system is beyond the scope of these construction rules. The Referee should be the final arbiter of what is considered normal, legal, licensed, or illegal in their setting. Remember, the rules will not be same in every star system and starship crews must be careful to ensure they comply with local regulations. In almost every case, nuclear weapons are restricted to military use only.



WEAPONS

Weapons may be installed in Turrets, Bays, or as Main Weapons. Not every weapon can be installed in all mounting systems at the same tech level, see the tables below for details.

Regular missiles cost Cr1200 each; smart missiles, Cr2500 each; nuclear missiles, Cr50000 each but are not normally available to civilians. Smart missiles are accurate, regardless of the gunner's skill. If they miss their target, they may make one more attempt to hit in the following round. Nuclear missiles are Intermediate weapons and cause 2D hits + 1 radiation hit. Ortillery missiles attack ground targets and cost Cr5000 per missile. See the Space Combat chapter for missile attack procedures.

A single **missile rack** holds 3 missiles, and each ton of cargo space modified as a missile magazine can accommodate 12 missiles. Loading a rack from nearby ordinary cargo takes the gunner a full space combat round. A pre-designated missile magazine allows automatic loading but prevents that space from being used for cargo when not carrying missiles.

Torpedoes are larger military-grade guided weapons. They pack an enormous punch, but are vulnerable to spoofing and point defense, like normal missiles. A torpedo launcher holds one torpedo, while torpedo bays fire salvoes of 12 torpedoes. Each torpedo costs Cr5,000, and each ton of cargo space can hold 6 torpedoes. Like missiles, pre-designated torpedo magazines allow for automatic reloading, but prevent the space from being used for cargo at other times.

Pulse lasers are ideal for point defense against incoming missiles. They are powerful but inaccurate, causing 2 hits on impact, but attacking at DM-2 to hit ordinary targets. See the Space Combat chapter for rules on point defense.

Mining lasers are a tool more than a weapon. Belters use these lasers to penetrate asteroids and reach valuable ores. Salvagers also use mining lasers to enter derelict ships or cut the ships up for parts. In a pinch, mining lasers can be used as a potent short-range weapon: they suffer DM-3 to all attack rolls when used as a weapon.

Sandcasters launch anti-laser sand canisters, described in the Ship Combat chapter. A single sandcaster holds 6 canisters and a single ton of cargo, or magazine, can hold 24 canisters. Sandcaster magazines can serve for sand canisters just as for missile magazines, and a single magazine can be used for both interchangeably, at an exchange rate of two canisters per missile.

Particle Beams blast enemies with beams of charged particles, causing both structural and radiation damage. This is the lightest military-grade energy weapon available, and many polities restrict its use to military and paramilitary ships.

Beam Lasers are weak but accurate lasers firing continuous beams.

Plasma Beams: are powerful beam weapons firing ionized plasma.

Fusion Beams: beam weapons emitting plasma undergoing a fusion reaction.

Gravity Lances fire shearing beams of gravitons which are exceedingly powerful, but attack rolls are affected by the hull configuration: Streamlined: DM-2, Distributed: DM+2.

Meson Cannons fire streams of exotic particles that decay inside the hull of the target ship. They are notoriously inaccurate, but when they hit, they can cause tremendous damage. Meson cannons consider armor as one level lower than usual, but are affected by meson shields.

Cherenkov Cannons fire a stream of particles that are moving faster than the speed of light in the local medium (tachyons) and can easily penetrate armor. Reduce target's armor rating by one category against this weapon.

Turret Weapons

Weapon	TL	Type	Hits	Cost	Notes
Missile Rack	6	Light	1D per missile	Mcr0.75	Nuclear missiles: Intermediate, 2D hits + 1 Radiation.
Pulse Laser	7	Light	2	Mcr0.5	DM-2 to hit; superior point defense
Sandcaster	7	Special	Special	Mcr0.25	
Particle Beam	8	Light	1 + 1 Radiation	MCr2	
Mining Laser	8	Light	1	Mcr0.15	DM-3 to hit
Beam Laser	9	Light	1	MCr1	
Plasma Beam	12	Light	3	MCr3	
Fusion Beam	14	Intermediate	3+1 Radiation	MCr4	
Gravitic Lance	16	Intermediate	4	MCr5	

Large Weapons

Large weapons are turret-mounted weapons, used almost exclusively by military ships. A single large weapon can fit in a triple turret, and takes up all the turret's space. These larger weapons are designed to penetrate low armor values and are commonly found on escorts and frigates. The following large weapons are available.

Large Weapons

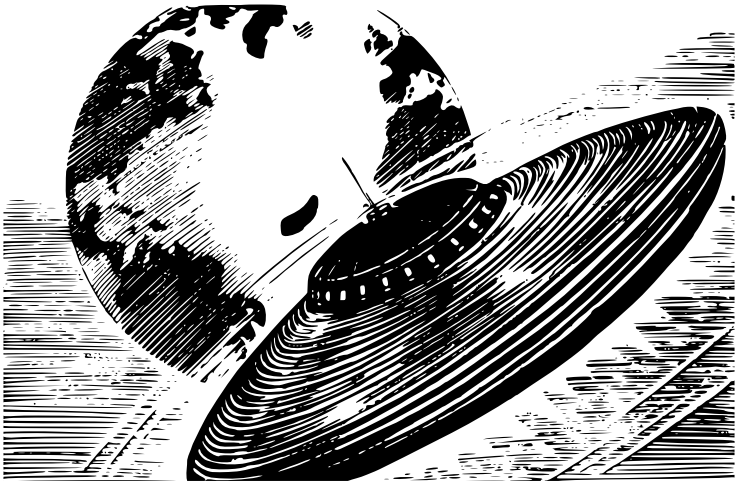
Weapon	TL	Type	Hits	Cost	Notes
Laser Cannon	10	Intermediate	2	MCr5	
Torpedo Launcher	10	Intermediate	3D	MCr4	Launches a single torpedo
Particle Cannon	11	Intermediate	2 + 1 Radiation	MCr10	
Plasma Cannon	12	Intermediate	3 + 1 Radiation	MCr15	
Fusion Cannon	13	Intermediate	4 + 1 Radiation	MCr20	
Gravitic Lance	16	Heavy	5	MCr25	

Bays

A bay weapon is a massive, military-scale gun emplacement designed for large-scale naval warfare. Each bay displaces 50 tons and uses 5 hardpoints. The following bay weapons are available.

Bay Weapons

Weapon	TL	Type	Hits	Cost	Notes
Mass Driver	8	Intermediate	2	MCr10	
Laser Cluster	8	Intermediate	2	MCr20	
Torpedo Bay	6	Intermediate	3D per torpedo	MCr20	Fires 5 torpedos
Particle Gun	9	Heavy	3 + 1 Radiation	MCr50	
Meson Gun	11	Heavy	4 + 1 Radiation	MCr150	
Fusion Gun	12	Heavy	4 + 1 Radiation	MCr100	
Gravitic Lance	15	Heavy	5	MCr125	
Cherenkov Cannon	16	Main Gun	5	MCr300	



MAIN GUNS

A large starship may carry massive, fixed-mounted, Main Guns in addition to its turrets and bays. A main gun displaces 1000 tons and requires ten hardpoints. There are three types of main guns, Particle Gun, Meson Gun, and Gravitic Disruptor. Note that the Meson Cannon and Gravitic Disruptor ignore armor.

Main Guns

Weapon	TL	Type	Hits	Cost	Notes
Particle Cannon	12	Main Gun	5 + 1 Radiation	MCr500	
Adv. Particle Cannon	14	Main Gun	5 + 2 Radiation	MCr1000	
Meson Cannon	12	Main Gun	4 + 1Radiation	MCr1500	DM-2 to hit
Adv. Meson Cannon	14	Main Gun	5 + 1 Radiation	MCr3000	No negative DM to hit
Gravitic Disruptor	15	Main Gun	6	MCr1250	
Cherenkov Projector	16	Main Gun	8	MCr3000	

SHIELDS

Shields are defensive energy systems that protect against specific attacks.

Shields

Shield	TL	Effect	Tons	Cost (MCr)
Meson Shield	12	Protects against Meson Weapons by halving the number of hits inflicted (round down) and by removing their special armor piercing characteristic. The shields also prevent the radiation damage from meson weapons.	50	60
Nuclear Damper	12	Halves fusion gun and nuclear missile hits (rounded down), removes automatic radiation hit from fusion and nuclear missile attacks.	50	50
Gravitic Field	13	Uses gravitic pulses to deflect incoming attacks, halving the hits from all weapons, but imposes a DM-1 on all outgoing attacks. At TL14, ignore the attack penalty.	100	100
Force Field	16	Ignores one attack, regardless of its power; this depletes the shield. Roll 1D on the following round. The Field regenerates on 4-6. Check each round until the Force Field regenerates.	200	300

Ship Crew

All vessels require a crew to operate and maintain the ship. A single person can run a 100-ton ship if they have the requisite skills, though a larger crew is advisable. Larger ships require larger crews.

Small Craft Crew

Each small craft requires a pilot, who can also fire the small craft's weapons. A dedicated gunner allows more complex gunnery actions, as given in the Space Combat chapter.

Ship Crews

Any ship requires a **pilot** to fly and steer the ship, as well as plot its course and calculate jumps. These tasks all require the Piloting skill. Military ships often have 3 pilots, working in four or eight hour shifts for constant monitoring of the flight controls and sensors, when not in jump space.

Some civilian ships and all military ships have a **sensor operator**, using the Computer skill. On smaller civilian ships, the pilot often fulfills this role, though this limits the utility of sensors in emergency or combat. All military ships carry a sensor operator, and sometimes at least three of them to ensure round-the-clock sensor monitoring.

All ships require one **engineer** per 35 tons of drives and power plant. Ships with less than 35 tons of drives and power plant, as well as small craft, do not require dedicated engineers, as their drives are small enough for the Pilot to handle during flight. However, an Engineer may be beneficial in ship combat. Engineers need the Engineering and Repair skills.

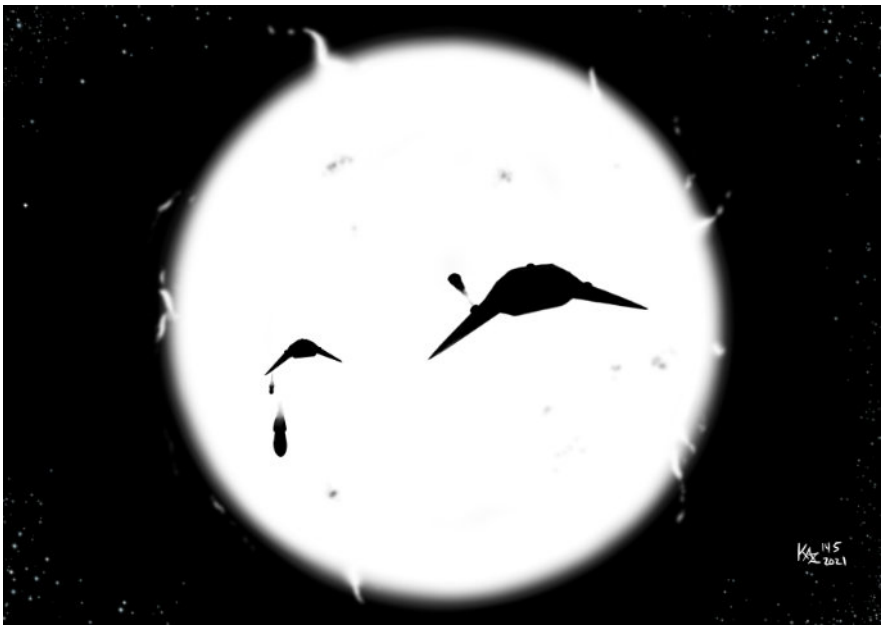
Any ship carrying passengers requires one or more **stewards**. See the discussion of Stewards in the Ship Revenue section.

Larger ships must have a **medic** to tend to the health of passengers and crew. Even ships without passengers often carry a medic to maintain the crew's health and handle medical emergencies. There must be one medic per 50 crew and passengers carried, including low passengers. Dedicated medbays require additional medical staff.

Each turret requires a **gunner** to operate it. Weapon bays require two gunners, while main guns require ten gun crewmembers each. Each shield requires one gunner. Ships with Model/2 or better electronics may automate turrets: one turret per Sensor DM bonus (including DM+0) can be automated. For example, a ship with sensor bonus DM+2 could automate three turrets; the computer has a Gunnery-1 skill. Weapons bays, main guns, and screens cannot be automated. In accordance with naval doctrine, most military craft prefer living gunners for all weaponry and defenses.

Military ships often carry **marines**, as well as flight crew – one pilot and one support crew per small craft on board. Larger civilian ships often have a purser and security staff aboard as well. Note that small craft pilots and service personnel are optional – many civilian starships do not have such dedicated personnel for their small craft.

All ships have a **captain**, though on smaller ships this is usually a part-time job performed by the pilot in addition to their regular duties. A dedicated captain is ideal for combat vessels and all-but-required on all military vessels.



ACCOMMODATIONS

The standard accommodations for ship crews are one **stateroom** per crew member. A stateroom displaces 4 tons and costs MCr0.5.

Military ships and more rugged freighters often institute a policy of **double occupancy**: housing two crew members or passengers in each stateroom. This does not increase the stateroom cost but is quite cramped. Highly skilled or senior civilian crew members and military officers usually get their own staterooms. Double occupancy for all crew except for the bridge crew or even for just the captain is standard practice on smaller military ships.

High and Middle Passage passengers require their own staterooms. Steerage class passengers always travel in double occupancy, unless the ship takes off with an odd number of Steerage class passengers, in which case the captain will organize a draw and one Steerage class passenger will have their own stateroom at no additional cost.

Cryotubes carry Low Passage passengers in cold sleep. Cyrotubes also serve as emergency transport and for carrying prisoners or captured lifeforms. A cryotube displaces 0.5 tons and costs MCr0.05.

Emergency cryotubes carry four persons each, who all share the same risky revival throw as noted in the Passage Section. They serve in emergencies, as well as for transporting large livestock. Most polities forbid carrying passengers in emergency cryotubes in non-emergency situations. Each displaces 1 ton and costs MCr0.1.

ADDITIONAL COMPONENTS

The following table lists various additional ship components.

Additional Components

Component	Tonnage	Cost (MCr)
Airlock, Additional	1	0.2
Armory (per 10 marines)	2	0.5
Breaching tube	3	3
Fuel processors (per 5 tons of fuel/day)	1	0.1
Fuel scoops	-	1
Galley	Included	Included
Medbay	Included	Included
Medlab (dedicated; per 2 patients)	8	2
Laboratory	4	4
Ship's locker	Included	Included
Small craft or vehicles	craft tonnage	As small craft
Escape pod (per person)	0.5	0.1
Mining Equipment	10	2
Probe drones (per 5 drones)	1	0.2
Repair drones	1% of hull	0.5/ton

Each ship is built with one standard **airlock** per 100 tons as part of the hull at no additional cost. Small craft do not come with airlocks standard. Additional airlocks are 1 ton and cost MCr0.2 each.

Military ships often carry an **armory**, providing military-grade small arms, armor, and equipment. Each 2-ton block of armory costs MCr0.5 and equips 10 marines.

A ship may use a **breaching tube** to breach and board a disabled enemy ship, that is, an enemy ship without an operable Maneuver Drive. A breaching tube takes up 3 tons of space and costs MCr3.

Fuel processors can refine unrefined fuel into refined fuel. A ton of fuel processing equipment can refine 5 tons of liquid hydrogen per day. Processors cost MCr0.1 per ton.

All streamlined ship hulls have built-in **fuel scoops** for collecting hydrogen fuel from gas giants. For ships with standard hulls, installing fuel scoops cost MCr1 and do not displace any tonnage.

Every ship has a **galley** (kitchen and lounge) and a **medbay**. These are included in stateroom tonnage and costs. A medbay counts as a medical facility for the purposes of healing wounds. A basic **medbay** accommodates up to 2 patients. A dedicated medlab, costing MCr2 and displacing 8 tons, represents better, hospital-grade care. **Medlabs** grant DM+1 to all Medicine tasks and allow doctors to perform Trauma Surgery without penalties for working in the field. Each additional 2 tons and MCr0.25 allocated allows another patient. Medlabs require one medic per 4 patients.

A **laboratory** is useful for scientific research, and comes in 4-ton blocks, each costing MCr4. A laboratory provides DM+1 to Science throws by the scientist using it. At the Referee's discretion, a laboratory is required for advanced or obscure scientific investigation.

Every ship includes a **ship's locker**, which contains various technical and survival supplies, space suits, and civilian small arms such as pistols and shotguns. We recommend that the Referee determine the exact contents of the ship's locker only when the players wish to access it. In any case, access to the ship's locker is usually controlled by biometric lock, keyed to the captain or other senior crew.

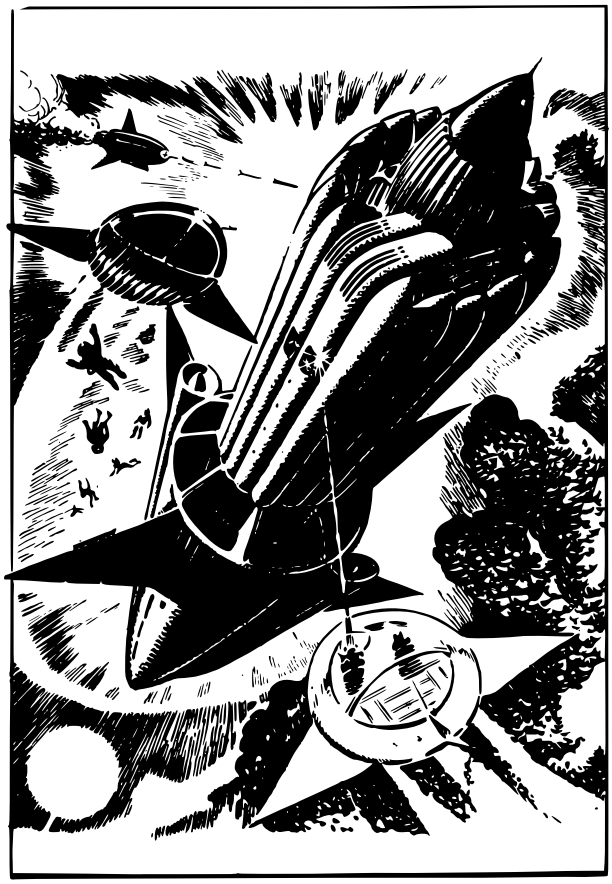
Ships may carry **small craft and vehicles**. The space allocated for a small craft or vehicle does not include space for spare parts or other maintenance equipment. All spares and equipment must be carried as cargo.

Most starships carry **escape pods**, which displace 0.5 tons and cost MCr0.1 per passenger or crew.

Mining equipment, displace 10 tons and cost MCr2, and allow a ship to mine asteroids. Each bay of mining equipment includes the mining drones and machinery to process 1Dx10 tons of asteroid per working day. The tonnage allocated also includes ore handling machinery, allowing the ship to take on ore and transfer it to the cargo bay.

Probe drones, displacing 1 ton and costing MCr0.2 per 5 drones, are designed for planetary surveying. The ship drops them from orbit in disposable re-entry capsules; they must be recovered manually. Alternatively, a probe drone in orbit can serve as a survey and communication satellite.

Repair drones displace 1% of the ship's hull and cost MCr0.5 per ton of drones. The complement of drones allows the ship to make one free repair attempt per space combat round. The drones have Repair 1 and Engineering 1 and can only perform or assist the Damage Control action. See the Space Combat chapter. Cost includes software.



Carried Vehicles

Starships may carry regular vehicles on board. The following table lists various transportable vehicles, their cost, and their size in ship tons. The tonnage listed includes space for access and basic maintenance equipment. Parts for maintenance need to be stored separately.

Carried Vehicles

Vehicle	Cost	Tonnage
4x4 Utility Vehicle	MCr0.04	4
Armored Personnel Carrier	MCr0.06	8
Bus	MCr0.035	12
Exploration Vehicle	MCr0.5	16
Ground Car	MCr0.006	3
Pick-Up Truck	MCr0.01	6
Tank	MCr0.3	20
Tripod	MCr3	24 (Folded)
Truck	MCr0.05	20
Walker	MCr0.5	8
Motorboat	MCr0.02	6
Patrol Boat	MCr0.1	30
Steamship	MCr0.72	50
Submersible	MCr1.7	100
Biplane	MCr0.02	8
Bomber	MCr1	30
Fighter, Propellor	MCr0.5	12
Fighter, Jet	MCr10	10
Helicopter	MCr0.25	8
Jet Liner	MCr5	60
Jump Jet	MCr0.5	20
Aircar	MCr0.15	4
Grav Barge	MCr0.5	16
Grav Carrier	MCr1	10
Grav Disc	MCr0.11	6
Grav Gunship	MCr2	16
Grav Liner	MCr10	80
Grav Orb	MCr0.02	2
Grav Skiff	MCr0.1	4
Speeder	MCr0.9	8

CARGO AND FINAL COST

The designer can now allocate any remaining tonnage to the ship's cargo bay.

Add up the cost of various components for the ship's final cost. Mass-produced ships enjoy a 10% discount to the final cost.

The question whether a ship is mass-produced is left to the Referee. Custom ships designed by players should not be mass-produced, as compared to standard models pre-designed by the Referee.

STARSHIP DESIGN EXAMPLE

Aaron is a Referee is looking for an interesting ship to give to his players since one of them, Jenny, rolled a Free Trader four times. This means that the player characters have a 40-year-old rustbucket of a trader with ten years of mortgage payments remaining. However, Aaron has a more interesting idea in mind: what if, in lieu of the trader, Jenny's character gets an old, decommissioned military transport? Unfortunately, after perusing the Common Space Craft chapter, he can't quite find a suitable ship. After cursing the Stellagama writers for such a glaring oversight, he decides to make one himself.

First, Aaron needs a concept for the ship. This would be an old light military transport. Not a warship by any means, but a light freighter used to fill holes in the wartime logistics networks and perform light transport duties. It wasn't intended for front-line space-combat duty but could defend itself when needed. This means something tougher than a trader, even if it falls within the typical trader-merchant tonnage range. Now that the war is over, the military has no need for so many old light transports. They decommission them, and then sell them on the open market. Jenny's character managed to get her hands on one through an old military contact. The ship's tech level is at the lower end of wartime military tech, as these ships were designed to be repaired anywhere, even on low-tech backwaters; therefore, Aaron puts the design at TL11.

A 300-ton hull will suffice: this is a light ship, after all, but not a fully-fledged bulk freighter. A 300-ton hull costs MCr30, but Aaron wants a ship that can go anywhere – a streamlined ship – so this increases the hull cost by 10%, putting it at MCr33. Since this is a ship that took enemy fire, like when landing at a hot LZ, armor is prudent; Light armor composed of TL10 Crystaliron will do the trick. This requires 6.5% of ship tonnage (19.5 tons) and costs 25% of the hull's cost (which is MCr33), that is MCr8.25.

Now, on to the engineering compartment. This Light Military Transport does not need high speeds, but it does need reasonably good legs; this means 2-G acceleration and Jump-2 capability. Aaron cross-references these performances on the Ship Drives table. He determines that he needs 4.5% of the ship's tonnage for the Jump Drive (13.5 tons) modified by TL for 1.2 for a total size of 16.2 tons, at a cost of MCr2.5 per ton, for a total cost of MCr40.5. The Maneuver Drive is 1.5% of the hull (4.5 tons) modified by 1.1 for TL11: the M-drive's total size is 5.0 tons, at a cost of MCr2.5 per ton (total MCr12.4). The Power Plant must be rated at 2, which requires 2.6% of the hull modified by 1.5 for TL11 (11.7 tons) at a cost of MCr2.5 per ton (MCr29.25). If the drives could be upgraded to TL12, the ship would gain 7.1 tons of additional cargo space. Aaron jots this down as a suggested player goal and adventure idea. Now, the engines need fuel. For a single Jump-2 capability, the ship needs 20% of its tonnage in liquid hydrogen fuel to form the jump bubble. That means 60 tons of jump fuel.

The ship needs a bridge. At 300 tons, the base 2% bridge displaces 6 tons, but starships have a 10-ton minimum. The bridge costs MCr0.5 per 100 tons of ship, so MCr1.5 for a 300-ton ship. Adjacent to the bridge is a computer and its associated electronics. Jump-2 capability requires a minimum Model/2 computer, but Aaron wants reasonable sensor capabilities; he chooses a Model/3 Computer, costing MCr3 and displacing 3 tons of electronics and providing a DM+1 to sensor checks.

On to the fun part – guns! As Aaron noted, this isn't a warship, so there is no need for heavy weapons, but a bunch of lasers and some defensive sandcasters will provide some protection in times of need. This is especially important as Aaron plans to introduce some pirates as the primary antagonists in his campaign, so the group will need some lasers to blast at them! He chooses to install three triple turrets. Each triple turret requires four tons, for 12 tons in total. Triple turrets cost MCr1 each themselves. One turret will have 3 pulse lasers – inaccurate but powerful and great for point-defense fire; each costing another MCr0.5, so, together with the turret, MCr2.5. The second turret will carry 3 sandcasters, with each costing MCr0.25. Together with the turret, this amounts to MCr1.75. Aaron decides that a military transport would have likely carried one more powerful weapon in the third triple turret. However, since the ship was decommissioned, the military scrappers removed the particle cannon (a Large weapon). The triple turret remains, for a cost of MCr1.